FUN-2.A.1 If a function is differentiable at a point, then it is continuous at that point. In particular, if a point is not in the domain of f, then it is not in the domain of f'.
FUN-2.A.2 A continuous function may fail to be differentiable at a point in its domain.

2-4 Differentiability and Continuity

Get Ready...

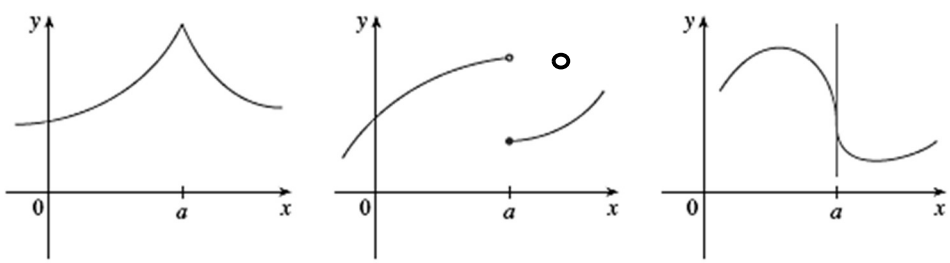
Definition of Continuity at a Point	A function f is **continuous at $x = c$** when three conditions are met: 1. 2. 3.

In essence, a function $f(x)$ is continuous at $x = c$ when you can draw the graph without lifting up the pencil or stopping.

Differentiability	• We say that a function $f(x)$ is **differentiable at $x = c$** if $f'(c)$ exists. (There is a derivative) • We say that a function $f(x)$ is **differentiable on an interval (a, b)** if $f'(c)$ exists for every point $x = c$ in the interval (a, b).

For a function $f(x)$ to be differentiable at $x = c$, two things MUST be true:
1. $f(x)$ must be **continuous** at $x = c$. 2. The derivatives from the left and right sides as x approaches c **must approach the same value**. (Graphically: the slope must approach the same value from both sides).

How can an equation (or the graph of an equation) fail to be differentiable?
1. the graph has a corner (or cusp) at $x = a$ 2. the function has a discontinuity at $x = a$ (_____, _____, _____) 3. the function has a vertical tangent at $x = a$

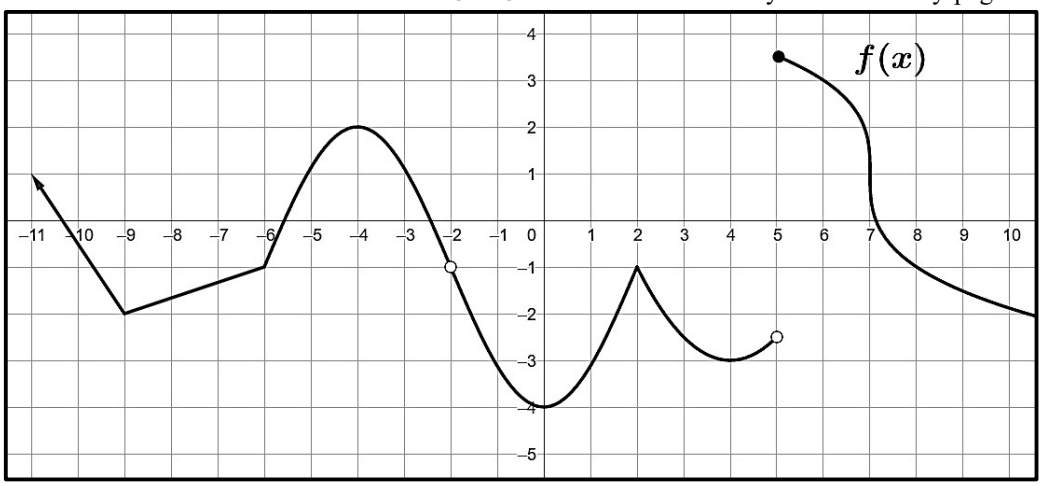


A corner A discontinuity A vertical tangent

Theorem	If f is differentiable at a, then f is continuous at a.

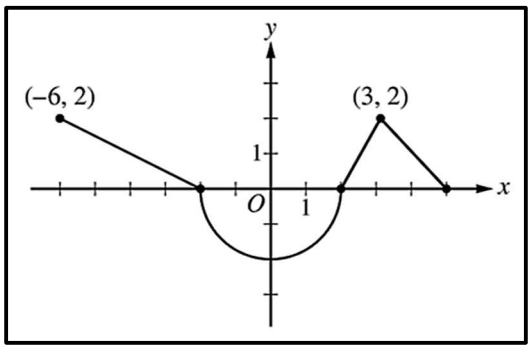
Higher Order Derivatives	If f is a differentiable function, then it's derivative, f' is a function, so f' may have a derivative of its own, denoted by $(f')' = f''$. This new function f'' is called the second derivative of f. Using Leibniz' notation, the second derivative of $y = f(x)$ is: $$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}$$

Example 1: Find all values of x where the function $f(x)$ at the right is not continuous. Classify each discontinuity.



Example 2: Find all values of x, with reasons, where the function $f(x)$ above is not differentiable.

Example 3: Consider the function $g(x)$ at the right. Find the following values, if possible.



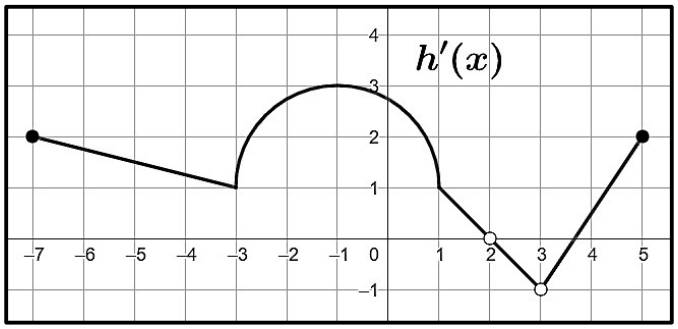
a. $g(-4)$ b. $g'(-4)$

c. $g(3)$ d. $g'(3)$

e. $g(-2)$ f. $g'(-2)$

g. $g'(2)$ h. $g'(4)$

Example 4: The graph of the function $h'(x)$ is at the right. Use the graph of $h'(x)$ to find the following, if possible.



a. $h'(-1)$ b. $h'(1)$

c. $h'(2)$ d. $h'(4)$

e. $h'(-5)$ f. $h''(-5)$

g. $h'(-3)$ h. $h''(-3)$

i. $h''(-1)$ j. $h''(4)$ k. $h''(2)$ l. $h''(1.17)$